

Mohammad Sharique Arshad

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SKILLS

- **Languages:** Embedded C, C, Python
- **Embedded:** ESP32, Arduino, GPIO, Timers, Embedded Firmware, Hardware Prototyping, PCB Design, Debugging & Testing
- **Communication & Tools:** LoRa, UART, SPI, I2C, WiFi, BLE, Proetus, Arduino IDE, KiCad, Github
- **Soft Skills:** Problem Solving, Adaptability, Team Collaboration, Quick Learner

TRAINING

AutomateX: Mastering PLC for Industrial Automation

Jun 2025 - Jul 2025

Summer Training

- Completed 50+ hours of industrial automation training covering PLC Programming, HMI, and SCADA.
- Developed a Tank Level Monitoring and Control System using Ladder Logic in LogixPro.
- Performed industrial process simulation and learned real-time automation techniques used in manufacturing.
- Worked with PLCs systems and continuous industrial process control system.

PROJECTS

Smart Helmet for Real-Time Accident Detection & Emergency Response([Link](#))

May 2025

- Engineered an ESP32-based smart helmet integrating MPU6050 IMU, SW420 vibration sensor, SIM800L GSM, NEO-6M GPS, LCD, and buzzer for live accident detection and emergency response.
- Implemented sensor fusion algorithms combining vibration and motion analysis to reduce false accident alerts by approximately 40%, improving detection accuracy and system reliability.
- Programmed automatic emergency notification to transmit GPS coordinates within 5 seconds of detecting a high-impact collision using GSM-based SMS communication.
- Integrated 6+ hardware modules using UART and I2C communication protocols, optimized embedded firmware for real-time operation, and validated complete hardware functionality through prototype testing.

Tech Stack: Embedded C, ESP32, MPU6050, SW420, SIM800L, NEO-6M GPS, UART, I2C, Arduino IDE

Farmezee – LoRa-Based Smart Irrigation & Environmental Monitoring System([Link](#))

Nov 2025

- Architected a solar-powered dual-node ESP32 IoT system for real-time environmental monitoring and automated irrigation using 868 MHz LoRa communication with a wireless range of 5-10 km (line-of-sight).
- Integrated BME280, soil moisture, and rain sensors to continuously monitor environmental conditions and automate irrigation based on configurable soil moisture thresholds.
- Developed automatic and manual irrigation control using relay-based pump switching, enabling on device OLED monitoring, audio alerts, and wireless data transmission between field and base stations.
- Developed a modular hardware prototype using ESP32 development boards and standard sensor modules, with a roadmap for migration to a custom PCB to support large-scale manufacturing.

Tech Stack: Embedded C, ESP32, LoRa (RYLR998 868 MHz), BME280, MQ135, UART, I2C, Arduino IDE

JalSetu – Autonomous IoT Boat for Water Quality Monitoring (SIH 2025)([Link](#))

Sep 2025

- Collaborated in a team to develop an autonomous ESP32-based IoT boat for autonomous water quality monitoring and flood-response applications.
- Integrated pH, turbidity, and waterproof temperature sensors along with GPS and ultrasonic sensors for water quality assessment, obstacle detection, and autonomous navigation.
- Implemented GSM-based communication to transmit 100+ sensor readings per day to a centralized web portal for remote monitoring and analysis.
- Developed a machine learning-based water quality classification system that analyzed sensor data to classify water as Safe or Unsafe based on environmental conditions associated with waterborne disease risk.

Tech Stack: ESP32, Embedded C, GSM, GPS, Machine Learning, IoT, Arduino IDE

CERTIFICATES

- PCB Basic Design Course | Altium Jan 2026
- Data Base Management System | NPTEL Sep 2025
- Embedded C Programming for Microcontrollers | Udemy Nov 2024
- Python Programming | Scalar Sep 2023

ACHIEVEMENTS

- 3rd place, Robo Sumo - Cognitia, LPU Feb 2025
- 2nd place, Robo Offroad Racing - Utkansh, NIT Jalandhar April 2025
- 1st place, Tech Fest - Pushpa Gujaral Science City, Jalandhar Sep 2024
- 1st place, Circuit Solving - Tech Marathon (SEEE), LPU Dec 2023

EDUCATION

- Lovely Professional University**
Bachelor of Technology
Electronics and Communication Engineering; CGPA: 7.47

Phagwara, Punjab
Aug 2023 - May 2027
- Bradford International School**
Intermediate | 69%

Patna, Bihar
Mar 2023
- Benhur School and College**
Matriculation | 84%

Allahabad, Uttar Pradesh
Mar 2020

